

Austin Baty

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Curriculum Vitæ

ORCID: 0000-0001-5310-3466

Education and Research Experience

University of Illinois Chicago

Assistant Professor

2023 -

UIC High Energy Nuclear Physics Group (CMS, ePIC Collaborations)

Rice University

Research Scientist

2023

Rice Academy Postdoctoral Fellow — Advisor: Prof. Wei Li

2019-2023

Rice Relativistic Heavy Ion Group (CMS, H1 Collaborations)

Massachusetts Institute of Technology

Ph.D. in Physics

2013-2019

Research Assistant — Advisor: Prof. Gunther Roland

MIT Relativistic Heavy Ion Group (CMS Collaboration)

Texas A&M University

B.S. in Physics and Mathematics, Summa Cum Laude, University Honors

2009-2013

Research Assistant — Advisor: Prof. Peter McIntyre

2012-2013

Texas A&M Accelerator Research Lab

Awards and Honors

2022 Global Young Scientist Summit ([GYSS](#)) attendee

2022

CMS Collaboration Award

2020

Given “for his development work on the heavy ion track reconstruction and research activity aiming to evolve the tracker data storage scheme.”

Best talk award at 2019 USA LHC Users’ Association meeting

2019

Rice University Academy Postdoctoral Fellowship

2019

MIT School of Science Fellowship

2013-2016

Research Highlights

- Analyzer for precision experimental extraction of the speed of sound in the quark-gluon-plasma, using a novel method of examining head-on ultracentral heavy ion collisions.
- Analyzer for measurement looking for QGP-like effects in single jets from pp collisions. Observes deviations from models which suggest new connections between nuclear and high-energy physics.
- Main analyzer of a precision Z boson measurement in PbPb collisions, published in PRL. Proposes a method to remove modeling uncertainties and biases in LHC measurements.
- Main analyzer of the CMS Collaboration’s 2015 charged particle nuclear modification factor paper. It has been cited over 300 times and had a figure featured in the CERN Courier.
- Author on a paper searching for collectivity in e^+e^- collisions using archived 30-year-old data from LEP. Provides a strong argument for open-data and data archival initiatives.
- Award-winning contributions to the CMS tracking detector, including leadership of the heavy ion tracking group for 7 years and a nearly 10x increase in heavy ion DAQ rate compared to 2012.

Leadership Positions

LPCC Heavy Ion Working Group Convener 2023 -

This working group provides a forum for the Heavy Ion community to discuss priority topics. As a convener I organize public forums and topics of discussion.

ePIC Collaboration Council representative for UIC 2023 -

I represent UIC's interests in high-level meetings used to dictate the organization and operation of the Collaboration.

Project Manager - CMS MTD ETL Nuclear Physics Consortium 2023

I coordinate a group of 5 institutions (KU, MIT, ORNL, Rice, UIC) during the construction of this \$18.4M timing detector that will be installed in CMS. I am responsible for the project being executed on-time and on-budget.

CMS Heavy Ion Heavy Flavor and Spectra group leader 2022 -

Responsibilities include arranging meetings and providing analysis advice, reviewing new analyses to maintain quality, and arranging priorities for new data collection.

CMS Raw' Data Format Task Force Leader 2020 - 2023

The group investigated new data formats to optimize event throughput. Leadership involved managing different skill sets, and communication to ensure proper integration. Ultimately resulted in a 35% increase in number of events gathered for the LHC Run 3.

CMS Heavy Ion Tracking Contact 2016 - 2022

I coordinated 5-10 people working on tracking software. Responsibilities include mentoring students, organization, troubleshooting, and facilitating collaboration with other groups.

Certifications

Certified Associate in Project Management (CAPM) 2023

A globally recognized credential from the Project Management Institute indicating training in professional project management. Skills tested include project leadership, ability to control the cost, schedule, and scope of large projects, and the ability to manage various stakeholder interests.

Other Experimental Experience

CMS Certified Language Editor 2018 -

Indicates strong technical writing skills, and proficiency with $\text{\LaTeX}$. I edit articles for the collaboration before they are submitted to a journal.

Track trigger contact for 2016 CMS pPb run 2016

I was in charge of the single-track trigger strategy. This involved managing a menu of triggers having different thresholds and data collection rates

Scientific service work

Reviewer for Physical Review C 2022 -

Advocate for Congressional funding of physics 2020 - 2021

I am a member of the contingent from the US LHC Users' Association that advocates for NSF and DoE funding. I set up meetings with members of Congress to update them about our scientific mission and remind them of the importance of their continued support.

Experimental Session Chair

2021

At the 'Opportunities of OO and pO collisions at the LHC' workshop hosted at CERN.

■■■■■ Mentoring

PhD advisor

Zheng Huang: Dec. 2023 -

Shirsendu Nanda: Dec. 2023 -

Nicholas Barnett: Aug. 2023 -

Mentoring as a postdoc

Parker Gardner (grad student): 2019 - 2023

■■■■■ UIC Service Work

Doctoral Preliminary Exam Committee Member

I have served on the following students' committees:

Ziyue Zhang - 9/22/2023

Clayton Bennett - 12/4/2023

Gavin Wilks - 12/19/2023

Publications

As a member of the CMS Collaboration, I am a listed author on over 800 publications. A full list can be found at [inspireHEP](#). Items here are ones I made notable contributions to. Being primary author indicates I was one of the main analyzers on the analysis, while secondary author indicates I provided support in the form of corrections, cross-checks, software, etc.. Paper editing contributions usually involved significant editing/rewrites of the paper.

As a primary author

- Sept. 2023 The CMS Collaboration. *Extracting the speed of sound in the strongly interacting matter created in ultrarelativistic nuclear collisions*. [CMS PAS HIN-23-003](#)
- Sept. 2023 The CMS Collaboration. *Pseudorapidity distributions of charged hadrons in PbPb collisions at $\sqrt{s_{NN}} = 5.36$ TeV*. [CMS PAS HIN-23-007](#)
- July 2023 The CMS Collaboration. *Observation of enhanced long-range elliptic anisotropies inside high-multiplicity jets in pp collisions at the LHC*. [CMS PAS HIN-21-013](#)
- July 2023 Baty, A., Gardner, P., and Li, W. *Novel observables for exploring QCD collective evolution and quantum entanglement within individual jets*. [Phys. Rev. C 107, 064908](#).
- Aug. 2021 The CMS Collaboration. *Constraints on the Initial State of Pb-Pb Collisions via Measurements of Z-Boson Yields and Azimuthal Anisotropy at $\sqrt{s_{NN}} = 5.02$ TeV*. [PRL 127 \(2021\) 102002](#).
- Dec. 2020 The H1 Collaboration. *Search for collectivity in e-p collisions with H1*. [Conference Note](#).
- Nov. 2019 Badea, A., et al. *Measurements of two-particle correlations in e^+e^- collisions at 91 GeV with ALEPH archived data*. [PRL 123 \(2019\) 212002](#).
- June 2019 Baty, A. *Study of Parton Energy Loss in Heavy Ion Collisions using Charged Particle Spectra Measured with CMS*. [Doctoral thesis](#).
- Aug. 2018 The CMS Collaboration. *Charged-particle nuclear modification factors in XeXe collisions at $\sqrt{s_{NN}} = 5.44$ TeV*. [JHEP 10 \(2018\) 138](#).
- April 2017 The CMS Collaboration. *Charged-particle nuclear modification factors in PbPb and pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV*. [JHEP 1704 \(2017\) 039](#).
- Jan. 2016 The CMS Collaboration. *Measurement of transverse momentum relative to dijet systems in PbPb and pp collisions at $\sqrt{s_{NN}} = 2.76$ TeV*. [JHEP 01 \(2016\) 006](#).
- June 2015 The CMS Collaboration. *Jet Fragmentation Function in pPb Collisions at $\sqrt{s_{NN}} = 5.02$ TeV and pp Collisions at $\sqrt{s} = 2.76$ and 7 TeV*. [CMS PAS HIN-15-004](#).
- Sept. 2013 Sooby E. et al. *Candidate molten salt investigation for an accelerator driven subcritical core*. [Journal of Nuclear Materials \(2013\) 440\(1\), 298–303](#).
- June 2013 Baty, A. *Molecular Dynamics Simulation of the Transport Properties of Molten Transuranic Chloride Salts*. [Undergraduate research thesis](#).

As a secondary author

- Dec. 2023 Chen, Y.-C., et al. *Long-range near-side correlation in e^+e^- collisions at 183–209 GeV with ALEPH archived data.* [Submitted to PRL](#).
- March 2023 Arslanok M., et al. *Hot QCD White Paper.* [arXiv:2303.17254](#).
- March 2023 Achenbach P., et al. *The Present and Future of QCD.* [As submitted to the 2023 NSAC Long Range Planning committee.](#)
- June 2022 Chen, Y., et al. *Jet energy spectrum and substructure in e^+e^- collisions at 91.2 GeV with ALEPH Archived Data.* [JHEP 06 \(2022\) 008](#).
- June 2021 The CMS Collaboration. *First measurement of large area jet transverse momentum spectra in heavy-ion collisions.* [JHEP 05 \(2021\) 284](#).
- May 2021 The CMS Collaboration. *Measurement of b jet shapes in proton-proton collisions at $\sqrt{s}=5.02$ TeV.* [JHEP 05 \(2021\) 054](#).
- April 2019 The CMS Collaboration. *Jet shapes of isolated photon-tagged jets in PbPb and pp collisions at $\sqrt{s_{NN}}=5.02$ TeV.* [PRL 122 \(2019\) 152001](#).
- Dec. 2018 The CMS Collaboration. *Observation of medium-induced modifications of jet fragmentation in Pb-Pb collisions at $\sqrt{s_{NN}}=5.02$ TeV using isolated photon-tagged jets.* [PRL 121 \(2018\) 242301](#).
- April 2018 The CMS Collaboration. *Pseudorapidity distributions of charged hadrons in proton-lead collisions at $\sqrt{s_{NN}}=5.02$ and 8.16 TeV.* [JHEP 01 \(2018\) 045](#).
- March 2016 The CMS Collaboration. *Correlations between jets and charged particles in PbPb and pp collisions at $\sqrt{s_{NN}} = 2.76$ TeV.* [JHEP 02 \(2016\) 156](#)

As the collaboration's paper editor

- Oct. 2023 The CMS Collaboration. *Measurements of the azimuthal anisotropy of charmonia in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV.* [JHEP 10 \(2023\) 115](#).
- Mar. 2023 The CMS Collaboration. *Multiplicity dependence of charm baryon and meson production in pPb collisions at 8.16 TeV.* [CMS-PAS-HIN-21-016](#).
- Mar. 2023 The CMS Collaboration. *Probing small Bjorken- x nuclear gluonic structure via coherent J/Ψ photoproduction in ultraperipheral PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV.* [Submitted to PRL](#).
- July 2022 The CMS Collaboration. *Probing charm quark dynamics via multiparticle azimuthal correlations in 5.02 TeV PbPb collisions.* [PRL 129, \(2022\) 022001](#).
- April 2022 The CMS Collaboration. *Correlations between multiparticle cumulants and mean transverse momentum in small collision systems with the CMS detector.* [CMS-PAS-HIN-21-012](#).
- Sept. 2021 The CMS Collaboration. *Observation of forward neutron multiplicity dependence of dimuon acoplanarity in ultraperipheral PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV.* [PRL 127, \(2021\) 122001](#).
- July 2020 The CMS Collaboration. *Measurement of quark- and gluon-like jet fractions using jet charge in PbPb and pp collisions at 5.02 TeV.* [JHEP 07 \(2020\) 115](#).

Presentations

Talks at larger conferences or invited talks are indicated by bold font.

- Nov. 2023 Parallel talk, *Extracting the speed of sound in the strongly interacting matter in ultrarelativistic nuclear collisions*. Nov. 21, 2023. 14th International Workshop on Multiple Parton Interactions at the LHC, Manchester, UK.
- Sept. 2023 Plenary talk, ***Overview of Recent CMS Results***. Sept. 4, 2023. Quark Matter 2023, Houston, Texas.
- July 2023 Invited Seminar, ***Observation of enhanced long-range elliptic anisotropies inside high-multiplicity jets in pp collisions at the LHC***. July 18, 2023. LHC Seminar, LHC Physics Center at CERN (LPCC), Geneva, Switzerland.
- June 2023 Parallel talk, *Status of DOE-NP ETL Project*. June 1, 2023. US CMS Annual Collaboration Meeting, Carnegie Mellon University, Pittsburgh, Pennsylvania.
- April 2023 Colloquium, *Jets and trillion-degree matter: studying the emergence of QCD phenomena*. April 5, 2023. University of Illinois Chicago Physics Colloquium, Chicago, Illinois.
- Mar. 2023 Invited plenary, ***Overview of Recent Experimental Electroweak Probes Results***. March 27, 2023. Hard Probes 2023, Aschaffenburg, Germany.
- Jan. 2023 Invited seminar, ***Jets and trillion-degree matter: studying QCD at multiple scales***. Jan. 26, 2023. Yale University Nuclear, Particle, and Astrophysics seminar, New Haven, Connecticut.
- Oct. 2022 Parallel talk, *Constraints on the initial state of heavy ion collisions at the LHC using electroweak bosons*. Oct. 14, 2022. Fall Meeting of the Texas Section of the APS, Houston, Texas.
- June 2022 Invited plenary, ***Overview of recent heavy ion results***. June. 29, 2022. CMS Week, CERN, Switzerland.
- April 2022 Parallel talk, ***Electroweak probes as tools for understanding initial state effects in heavy ions with CMS***. April. 7, 2022. Quark Matter 2022, Krakow, Poland.
- Feb. 2022 Colloquium, *Trillion Degree Matter: Probing the emergence of the quark-gluon plasma*. Feb. 3, 2022. Vanderbilt University Department of Physics and Astronomy Colloquium.
- Feb. 2022 Invited seminar, *Trillion Degree Matter: Probing the emergence of the quark-gluon plasma*. Feb. 1, 2022. Rice University Nuclear and Particle Physics Seminar.
- Aug. 2021 Parallel talk, *Measurement of charged particle multiplicity distributions in DIS at HERA and its implication to entanglement entropy of partons*. Aug. 5, 2021. 14th Quark Confinement and the Hadron Spectrum Conference.
- Jan. 2021 Parallel talk, ***Measurement of initial stages via color neutral probes in pPb and PbPb***. Jan. 14, 2021. Initial Stages 2021.
- Oct. 2020 Parallel talk, *New constraints of initial states in PbPb collisions with Z boson yields and azimuthal anisotropy at 5.02 TeV*. Oct. 29, 2020. APS Division of Nuclear Physics fall meeting (DNP 2020).

- Oct. 2020 Invited talk, *Future opportunities for exploring collectivity using extra small systems*. Oct. 28, 2020. Snowmass EF07 meeting: High Density QCD in Small Collision Systems
- June 2020 Parallel talk, *Probing initial and final state effects with Z bosons in PbPb and Drell-Yan in pPb with the CMS detector*, June 3rd, 2020. Hard Probes 2020.
- Nov. 2019 Invited talk, *Electroweak Boson production in Heavy-Ion Collisions from CMS*, November 11th, 2019. A Workshop on Heavy Flavor and Dilepton Production in Relativistic Heavy-Ion Collisions (HeFe2019), University of Science and Technology of China, Hefei, China.
- Nov. 2019 Parallel talk, *Collectivity of Heavy Flavor Hadrons in pp and pPb Collisions with the CMS Detector*, November 5th, 2019. Quark Matter 2019, Wuhan, China.
- Oct. 2019 Parallel talk, *Using charged particles to study jet quenching in heavy ion collisions with CMS*, October 17, 2019. US LHC Users Association Annual Meeting 2019, Rice University, Houston, Texas.
- Oct. 2019 Invited seminar, *Searching for collectivity in small systems using e^+e^- data from ALEPH*, October 1, 2019. LBL Heavy Ion Tea Seminar, Berkeley National Lab, Berkeley, California.
- June 2019 Plenary talk, *Searches for the ridge in DIS and e^+e^-* , June. 27, 2019. Initial Stages 2019, Columbia University, New York City, New York.
- June 2019 Invited talk, *Studies of archived e^+e^- and HERA data*, June. 4, 2019. 2019 RHIC & AGS Annual Users' Meeting, Brookhaven National Lab, Brookhaven, New York.
- Feb. 2019 Invited talk, *Exploring charged particle production in the quark-gluon plasma: from e^+e^- to PbPb*, Feb. 7, 2019. CU Boulder, Boulder, Colorado.
- Dec. 2018 Invited talk, *Measurements of two-particle correlations in e^+e^- collisions at 91 GeV with ALEPH archived data*, Dec. 11, 2018. 10th International Workshop on Multiple Parton Interactions at the LHC, Perugia, Italy.
- Oct. 2018 Parallel talk, *Charged Particle Nuclear Modification Factors in pPb, PbPb, and XeXe Collisions with CMS*, Oct. 2, 2018. Hard Probes 2018, Aix-Les-Bains, France.
- May 2018 Parallel talk, *Charged Particle Nuclear Modification Factors in pPb, PbPb, and XeXe Collisions with CMS*, May 15, 2018. Quark Matter 2018, Venice, Italy.
- Sept. 2017 Parallel talk, *Boson+Jet Correlations and Boson-Tagged Jet Substructure in pp and PbPb collisions at 5.02 TeV with CMS*, Sept 20, 2017. Initial Stages 2017, Krakow, Poland
- Sept. 2016 Parallel talk, *Charged particle nuclear modification factor in PbPb at 5.02 TeV with CMS*, Sept 25, 2016. Hard Probes 2016, Wuhan, China
- July 2016 Invited talk, *Jet and Hadron R_{AA}* , July 27, 2016. 4th Heavy-Ion Jet Workshop, Paris, France
- May 2016 Parallel talk, *Charged-particle production in PbPb and pp collisions at 5 TeV with CMS*, May 24, 2016. Initial Stages 2016, Lisbon, Portugal
- Sept. 2015 Parallel talk, *Fragmentation patterns of jets in pPb collisions in CMS*, September 28, 2015. Quark Matter 2015, Kobe, Japan